

The GC - report

January 2025

1 Group name

The name of our project group in this course is "The GC".

1.1 Participant list

Name	Email	Active Dates
Åke Lindström	ake.lindstrom.2042@student.uu.se	3/12 2024 – 17/1 2025
Viktor Liljenberg	viktor.liljenberg.6239@student.uu.se	3/12 2024 – 17/1 2025
Haoran Wang	haoran.wang.1103@student.uu.se	3/12 2024 – 17/1 2025
Leo Stark	leo.johansson-stark.2628@student.uu.se	3/12 2024 – 17/1 2025
Hiro Andersson	hiro.andersson.0326@student.uu.se	3/12 2024 – 17/1 2025
Emilio Funes	emilio.funes.3811@student.uu.se	3/12 2024 – 17/1 2025

2 Quantification

We started planning and working on the project the 3rd of December.

- 360 timmar
- 92 git commits
- 6 issues
- 1 pull request
- Number of sprints, their start and end dates
- Total number of new lines of C code written excluding tests and preexisting code
- Total number of lines of test code
- Total number of lines of "script code" (e.g., make files, Python scripts for generating test data, etc.)

- Total number of hours worked by the team.
- Total number of git commits
- Total number of pull requests
- Total number of GitHub issues

3 Process

3.1 Inception

For this project, we intended to use Scrum as our development process. However, early on in the project we chose to focus on understanding the task at hand and getting started with the code. As a result, learning and implementing Scrum was put to the side. When working on the project we realized that our development process more aligned with the Kanban framework and decided to use that for the rest of the project instead.

To learn about Kanban we looked up various specifications on how it was supposed to work and tried to apply it to our project.

3.2 Implementation

We used Github Projects as a kanban board to assign tasks within the group. We did however never assign a coach to keep track of overloading in the project which led to some students feeling overwhelmed with work on different tasks. Our project board was set up with different tabs corresponding to the different stages of development. Each work item started off in the “TODO” tab and was eventually moved through the tabs “In progress”, “Testing” and “Done” during development.

We never defined proper sprints which came with both advantages and disadvantages. Having no sprints made it easy to get started, as our planning could be more flexible. It allowed different work pairs within the group to take on and complete tasks independently of others. It also made it easy to reprioritize tasks when needed. However, lack of defined sprints made it easier to postpone work. Proper sprints could have made sure that we worked more continuously during the whole span of the project and might have made it easier for us to hit our own soft deadline.

Our planning mostly took place in meetings, both in-person and virtually. We started off with discussing the order in which the different parts of the code would be completed and split the different parts up into different tasks that were assigned within the group. We also figured out some of the bigger design choices and set a soft deadline for when we wanted our code to be finished. The order of task completion and general design choices was overall followed but we

unfortunately missed our soft deadline. Partly due to underestimating our ability to work during the Christmas break but also due to communication issues within the team, leading to some students feeling stressed and overworked. We had no specific strategy to make sure we got work done during Christmas break, resulting in little progress being made during that time and an uneven workload on our members. Another contributing factor to missing the soft deadline was a tendency to underestimate how much time some of the tasks would consume. Parts of the garbage collector turned out to be more complicated than initially thought and it made it harder to accurately plan ahead.

Having one person be an agile coach is something we would do differently if we were to start over. Having someone focusing on coordinating the team might have improved the work balance within the team and how we communicated. Regular meetings could also help with solving our communication problems. Another improvement would be to have more deadlines and be more specific and detailed with our planning, to make sure that we get more work done earlier on in the project.

The use of a project board was a great tool for keeping track of what needed to be done and what everyone was working on. Although how well we used it varied during the project, it is still absolutely something worth trying again.

4 Use of Tools

We used git for version control and stored everything on Github. We could easily break up the work into different branches depending on what we were working on and whether it was still in development. We used discord as our main form of communication and often worked from home. Visual Studio Codes liveshare tool was useful when coding together from home, and we used it for pair programming during the project.

We made use of githubs project tool for assigning different tasks to people in the group and marking them as done or not. Getting the group to make use of this tool worked somewhat during the first days of the project but we mostly resorted to assigning tasks during the group meetings.

5 Communication, Cooperation and Coordination

We communicated mostly through discord with each other during the project. We had two meetings with our TA during the project where we received feedback on our work and we could set goals for ourselves.

In the beginning of the project we split our group into pairs and each pair was assigned a different task on the project that we could loosely work on independently of other pairs. Sometimes, it was necessary for two pairs to work together because two tasks depended too much on each other. In order to keep every pair synchronized we planned group meetings. Much time was lost during the break around Christmas and new years eve but we tried to work together from home between the holidays. Some people in the group had to work more than others and it was hard to find dates where everyone in the group could have a meeting together.

6 Work Breakdown Structure

In the beginning of the course we had a deadline that said we should have finished the code for the project on 7 January when we had our TA-meeting. We were not done on that date but we had finished most of the project. In the beginning of the project it was hard to understand how much time something would take to implement and what should be implemented first. The structure of the header was planned to be done fairly early during the project, but it was more complicated than what was planned for.

We split the project group into work pairs and assigned different tasks. Using pair programming went fairly well. Some pairs unfortunately sometimes had more work than others, so we had to rearrange groups and sometimes have several pairs work on one task.

7 Reflection

We are fairly sure in implementing the right thing. The project group regularly followed and read the instructions during group meetings. We are not sure however that the implementation is as effective as it should have been because of lack of testing due to our limited amount of time. We are not sure in how many spaces that should be active or inactive at the same time.

We were overall not satisfied with our project process. The choice of development process was not clearly defined and we did not have a plan for solving it during the beginning of the project. Due to lack of communications some people ended up not knowing what tasks to do some days and other members having too much work. We rated our satisfaction of the process as 3/7, because we got a finishing product in the end.

The finishing product did pass all of our own written tests in the end and we even had functionality for pointers into object which were not needed for the project. We are satisfied with finishing this fairly hard project and give it

a rate of 5/7.

We give the quality assurance a 6/7 for passing all tests and having good code coverage. The garbage collector could be more efficient and should take care of secure and unsecure pointers, which it doesn't in our implementation.

The biggest win the working on this project is getting the experience to work together in a large group on a project and also delivering a finishing product.

The biggest fail of this project was our planning. We did not have a clear choice of development process and had a hard time creating meetings that everyone could attend.